

Due Date: 04.18

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Solution:

a)

$$H(z) = 1 + z^{-1} + 7z^{-6}$$

Obviously, the only poles of $H(z)$ will be at $z = 0$, so there is no feedback term; the filter is FIR. We can verify this by taking the inverse z-transform:

$$h[n] = \delta[n] + \delta[n - 1] + 7\delta[n - 6]$$

b)

$$H(z) = \frac{z^2 + 3z + 2}{z + 1}$$

Expanding the numerator we can see there is pole-zero cancellation:

$$H(z) = \frac{(z + 1)(z + 2)}{z + 1} = z + 2$$

Much like in part a, we can take the inverse z-transform and see that it is an FIR filter:

$$h[n] = \delta[n + 1] + 2\delta[n]$$

c) The same pole-zero cancellation appears as in part b:

$$H(z) = \frac{z + 1}{(z + 1)(z + 2)} = \frac{1}{z + 2} = \frac{z^{-1}}{1 + 2z^{-1}}$$

This has a pole at $z = -2$, and is therefore IIR. This can be seen by taking the inverse z-transform:

$$h[n] = (-2)^{n-1}u[n - 1]$$

or a similar infinite-length result if the system is left-sided, making it IIR.

d)

$$h[n] = 2^{-n}(u[n] - u[n - 200])$$

Since the impulse response contains $(u[n] - u[n - 200])$, which is a rectangular window, the impulse response is finite in length, making it FIR.

e)

$$y[n] - \frac{1}{2}y[n - 1] = x[n]$$

Taking the z-transform of both sides:

$$Y(z) - \frac{1}{2}Y(z)z^{-1} = X(z)$$

Solving for $H(z)$:

$$H(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

Taking the inverse z-transform:

$$h[n] = \left(\frac{1}{2}\right)^n u[n]$$

or a similar infinite-length result if the system is left-sided, making it IIR.

f)

$$y[n] = \frac{1}{3}(x[n-1] + x[n] + x[n+1])$$

By inspection, this can be decomposed as:

$$y[n] = x[n] * \frac{1}{3}(\delta[n-1] + \delta[n] + \delta[n+1])$$

Because the system is LSI, this results in:

$$h[n] = \frac{1}{3}(\delta[n-1] + \delta[n] + \delta[n+1])$$

This impulse response is finite in length, making this a FIR system.

Grading: 30 points

- +2 Correct answer each part
- +3 Correct justification each part

1

Solution:

(a) Since the response is even symmetric, it is GLP.

$$\begin{aligned} H_d(\omega) &= 2 + e^{-j\omega} + 2e^{-j2\omega} \\ &= e^{-j\omega} (1 + 2(e^{j\omega} + e^{-j\omega})) \\ &= e^{-j\omega} (1 + 4\cos\omega) \end{aligned}$$

$$M = 1, \alpha = 0, R(\omega) = 1 + 4\cos\omega.$$

$\alpha = 0$ but $R(\omega) = 1 + 4\cos\omega$ crosses 0 at some ω , introducing π jumps in the phase, so this is not linear phase.

(b) We can decompose the response into the sum of an even symmetric and an odd symmetric response:
 $\{1, 2, 3\} = \{2, 2, 2\} + \{-1, 0, 1\}$

$$\begin{aligned} H_d(\omega) &= (2 + 2e^{-j\omega} + 2e^{-j2\omega}) + (-1 + e^{-j2\omega}) \\ &= (2e^{-j\omega}(1 + e^{j\omega} + e^{-j\omega})) + (e^{-j\omega}(-e^{j\omega} + e^{-j\omega})) \\ &= (2e^{-j\omega}(1 + 2\cos\omega)) + (e^{-j\omega}(-2j\sin\omega)) \\ &= e^{-j\omega}(2 + 4\cos\omega - 2j\sin\omega) \end{aligned}$$

Since $A(\omega)$ is not real, it is not a GLP.

(c)

$$\begin{aligned}H_d(\omega) &= e^{-j\omega} (3 + j(-e^{j\omega} + e^{-j\omega})) \\&= e^{-j\omega} (3 + j(-2j \sin \omega)) \\&= e^{-j\omega} (3 + 2 \sin \omega)\end{aligned}$$

$$M = 1, \alpha = 0, R(\omega) = 3 + 2 \sin \omega.$$

Since $\alpha = 0$ and $R(\omega) = 3 + 2 \sin \omega > 0$ for all ω , this is linear phase.

(d) Similar to the previous one, but here due to the different sign, we have

$$H_d(\omega) = e^{-j\omega} (3 + 2j \cos \omega),$$

so it is not a GLP.

(e)

$$\begin{aligned}H_d(\omega) &= e^{-2j\omega} (1 + e^{j2\omega} - e^{-j2\omega} + e^{j\omega} - e^{-j\omega}) \\&= e^{-2j\omega} (1 + 2j \sin(2\omega) + 2j \sin(\omega))\end{aligned}$$

Since there are complex terms in the function being multiplied to the complex exponential, this is not GLP.

(f)

$$\begin{aligned}H_d(\omega) &= e^{-j\omega} (e^{j\omega} - e^{-j\omega}) \\&= e^{-j\omega} (2j \sin(\omega)) \\&= 2 \sin(\omega) e^{j(-\omega + \frac{\pi}{2})} \\&= -2 \sin(\omega) e^{j(-\omega - \frac{\pi}{2})}\end{aligned}$$

$$M = 1, \alpha = \frac{\pi}{2}, R(\omega) = 2 \sin \omega, \text{ or } M = 1, \alpha = \frac{\pi}{2}, R(\omega) = -2 \sin \omega.$$

Since $\alpha = \frac{\pi}{2}$, this is not linear phase.

(g)

$$\begin{aligned}H_d(\omega) &= e^{j\frac{\pi}{2}} e^{-j\frac{3}{2}\omega} \left(e^{j\frac{\omega}{2}} + e^{-j\frac{\omega}{2}} + 2e^{j\frac{3\omega}{2}} + 2e^{-j\frac{3\omega}{2}} \right) \\&= e^{j\frac{\pi}{2}} e^{-j\frac{3}{2}\omega} \left(2 \cos\left(\frac{\omega}{2}\right) + 4 \cos\left(\frac{3\omega}{2}\right) \right) \\&= \left(2 \cos\left(\frac{\omega}{2}\right) + 4 \cos\left(\frac{3\omega}{2}\right) \right) e^{j(-\frac{3}{2}\omega + \frac{\pi}{2})}\end{aligned}$$

$$M = \frac{3}{2}, \alpha = \frac{\pi}{2}, R(\omega) = 2 \cos\left(\frac{\omega}{2}\right) + 4 \cos\left(\frac{3\omega}{2}\right).$$

Since $\alpha = \frac{\pi}{2}$, this is not linear phase.

Grading: 28 points

- +4 points each part

Solution:

The Direct Form II block diagram has the following LCCDE:

$$y[n] = x[n] + \frac{1}{3}x[n-1] - \frac{1}{4}x[n-2] - 3y[n-1] + \frac{1}{2}y[n-2]$$

$$y[n] + 3y[n-1] - \frac{1}{2}y[n-2] = x[n] + \frac{1}{3}x[n-1] - \frac{1}{4}x[n-2]$$

To find the transfer function, take the z-transform of the LCCDE:

$$Y(z) + 3Y(z)z^{-1} - \frac{1}{2}Y(z)z^{-2} = X(z) + \frac{1}{3}X(z)z^{-1} - \frac{1}{4}X(z)z^{-2}$$

$$Y(z)(1 + 3z^{-1} - \frac{1}{2}z^{-2}) = X(z)(1 + \frac{1}{3}z^{-1} - \frac{1}{4}z^{-2})$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + \frac{1}{3}z^{-1} - \frac{1}{4}z^{-2}}{1 + 3z^{-1} - \frac{1}{2}z^{-2}}$$

Grading: 22 points

- +11 Correct transfer function
- +11 Correct difference equation
- -1 each minor error

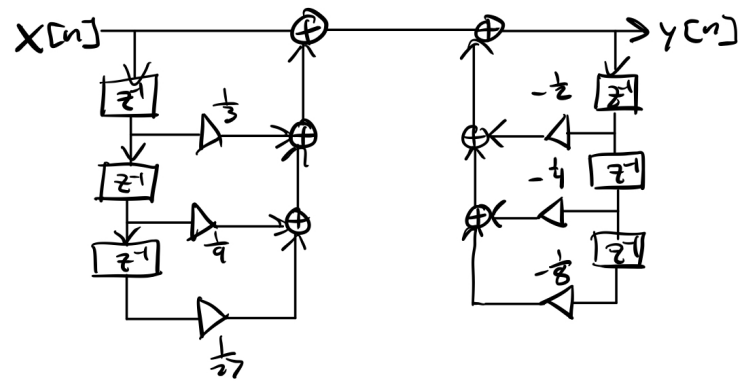
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Solution:

Grading: 20 points

- +10 Correct direct form 1
- +10 Correct direct form 2
- -1 each minor error

Direct Form I:



Direct Form II:

